RAJALAKSHMI ENGINEERING COLLEGE
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AD23532 – PRINCIPLES OF DATA SCIENCE
EXPERIMENT 6A: EXPLORATORY DATA ANALYSIS

QUESTION:
Perform Exploratory Data Analysis (EDA) on the Netflix Titles dataset to analyze content distribution, release trends, duration, genres, directors, actors, countries, ratings, and descriptions. Use statistical analysis and visualizations to identify patterns and insights in Netflix movies and TV shows.

AIM:
To perform Exploratory Data Analysis on the Netflix Titles dataset using Python by examining content types, ratings, release years, countries, durations, genres, directors, actors, and textual descriptions, and visualizing the findings using suitable graphs and charts.

ALGORITHM:
1. Import the required Python libraries: Pandas, NumPy, Matplotlib, Seaborn, and WordCloud.
2. Load the Netflix Titles dataset into a Pandas DataFrame and examine its shape, columns, data types, missing values, and summary statistics.
3. Analyze the distribution of Movies and TV Shows, content ratings, and countries using count plots and bar charts.
4. Analyze release years and date-added trends to identify the years with the highest content releases and Netflix content growth.
5. Examine movie durations and TV show seasons using histograms and bar charts.
6. Identify the top 10 genres, directors, and actors appearing most frequently on Netflix.
7. Generate a word cloud to identify common words in content descriptions.
8. Analyze monthly content additions, compare movie and TV show production by country, and examine ratings across content types.
9. Identify the oldest and newest titles and create a genre co-occurrence heatmap to study relationships between genres.
10. Interpret the visualizations and summarize the major findings from the dataset.

PROGRAM & OUTPUT:

```python
# Name: Shyam M (241801267)
# Section 1: Import Libraries and Load Dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
from itertools import combinations
from collections import Counter

df = pd.read_csv("/content/netflix_titles.csv")

print("Dataset Shape:", df.shape)
print("\nColumns:", df.columns.tolist())
print("\nData Types:\n", df.dtypes)
print("\nMissing Values:\n", df.isnull().sum())
print("\nSummary Statistics:\n", df.describe(include="all"))
```

```
Dataset Shape: (8807, 12)

Columns: ['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration

Data Types:
 show_id         object
type             object
title            object
director         object
cast             object
country          object
date_added       object
release_year      int64
rating           object
duration         object
listed_in        object
description      object
dtype: object

Missing Values:
 show_id            0
type                0
title               0
director         2634
cast              825
country           831
date_added         10
release_year        0
rating              4
duration            3
listed_in           0
description         0
dtype: int64
```

Summary Statistics:

	show_id	type	title	director	cast
count	8807	8807	8807	6173	7982
unique	8807	2	8807	4528	7692
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough
freq	1	6131	1	19	19
mean	NaN	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN

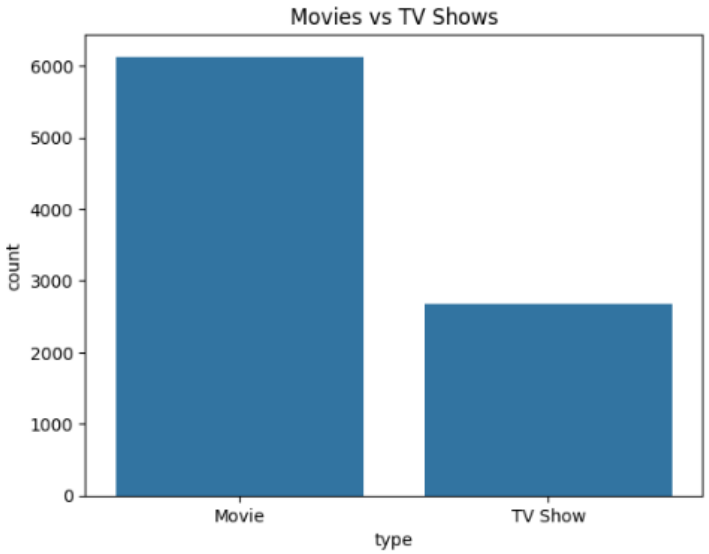
	country	date_added	release_year	rating	duration
count	7976	8797	8807.000000	8803	8804
unique	748	1767	NaN	17	220
top	United States	January 1, 2020	NaN	TV-MA	1 Season
freq	2818	109	NaN	3207	1793
mean	NaN	NaN	2014.180198	NaN	NaN
std	NaN	NaN	8.819312	NaN	NaN
min	NaN	NaN	1925.000000	NaN	NaN
25%	NaN	NaN	2013.000000	NaN	NaN
50%	NaN	NaN	2017.000000	NaN	NaN
75%	NaN	NaN	2019.000000	NaN	NaN
max	NaN	NaN	2021.000000	NaN	NaN

	listed_in
count	8807
unique	514
top	Dramas, International Movies
freq	362
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN

	description
count	8807
unique	8775
top	Paranormal activity at a lush, abandoned prope...
freq	4
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN

```
# Name: Shyam M (241801267)
# Section 2: Movies vs TV Shows

sns.countplot(data=df, x="type")
plt.title("Movies vs TV Shows")
plt.show()
```




```
# Name: Shyam M (241801267)
# Section 3: Ratings and Countries

df["rating"].fillna("Unknown", inplace=True)

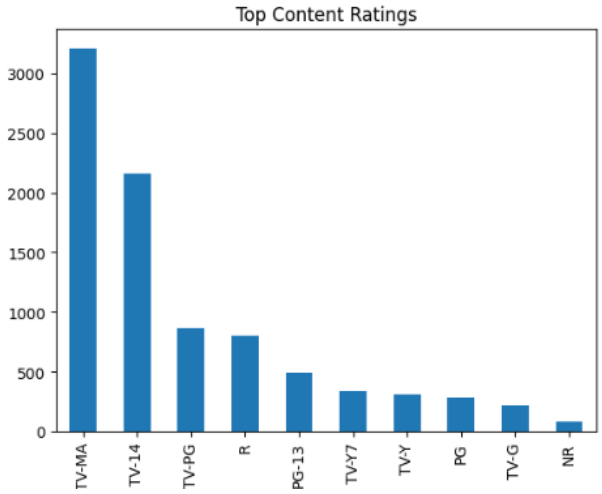
df["rating"].value_counts().head(10).plot(kind="bar")
plt.title("Top Content Ratings")
plt.show()

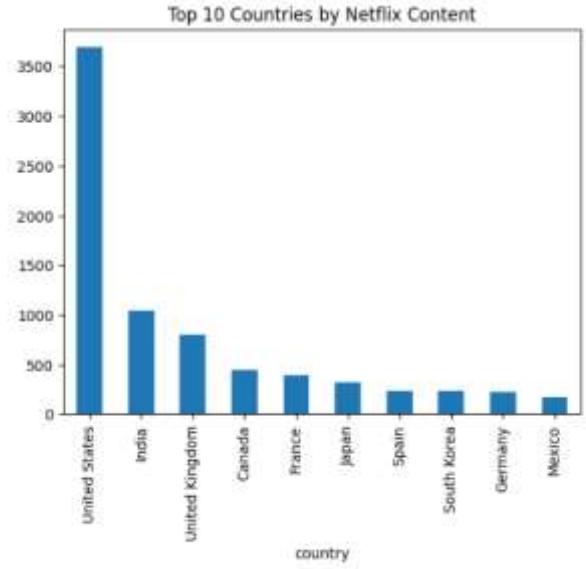
df["country"].dropna().str.split(", ").explode().value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Countries by Netflix Content")
plt.show()
```

```
/tmp/ipykernel_4296/2340315183.py:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignm
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].meth

  df["rating"].fillna("Unknown", inplace=True)
```

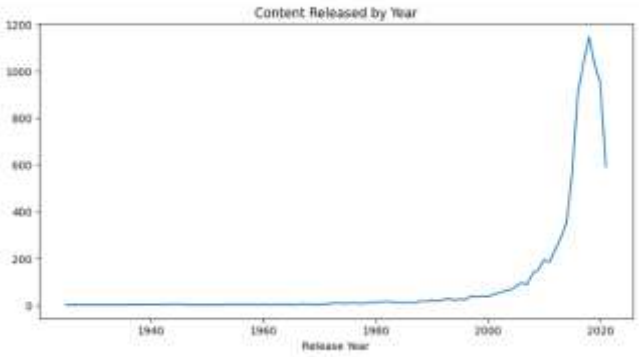


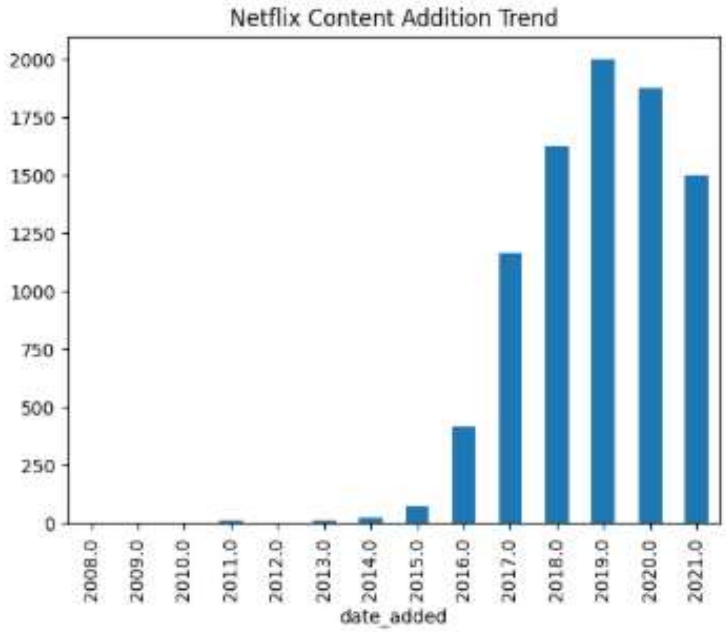



```
# Name: Shyam M (241801267)
# Section 4: Release Year and Content Growth

df["release_year"].value_counts().sort_index().plot(figsize=(10, 5))
plt.title("Content Released by Year")
plt.xlabel("Release Year")
plt.show()

df["date_added"] = pd.to_datetime(df["date_added"], errors="coerce")
df["date_added"].dt.year.value_counts().sort_index().plot(kind="bar")
plt.title("Netflix Content Addition Trend")
plt.show()
```





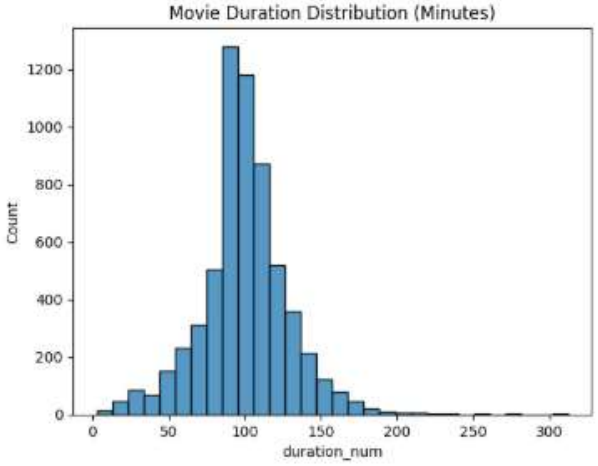

```
# Name: Shyam M (241801267)
# Section 5: Movie Duration and TV Show Seasons

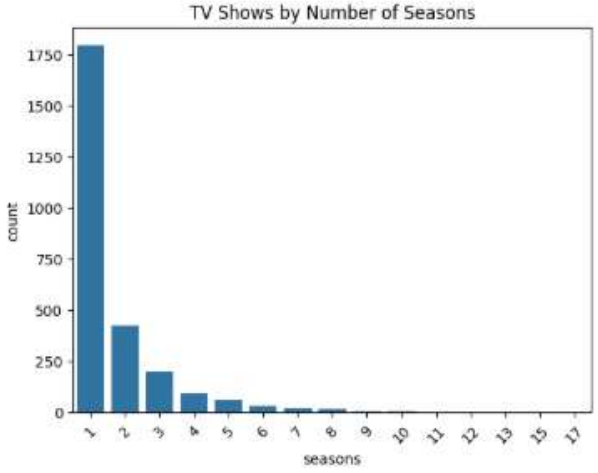
movies = df[df["type"] == "Movie"].copy()
movies["duration_num"] = pd.to_numeric(
    movies["duration"].str.extract(r"(\d+)")[0],
    errors="coerce"
)

sns.histplot(movies["duration_num"].dropna(), bins=30)
plt.title("Movie Duration Distribution (Minutes)")
plt.show()

shows = df[df["type"] == "TV Show"].copy()
shows["seasons"] = pd.to_numeric(
    shows["duration"].str.extract(r"(\d+)")[0],
    errors="coerce"
)

sns.countplot(data=shows, x="seasons")
plt.title("TV Shows by Number of Seasons")
plt.xticks(rotation=45)
plt.show()
```



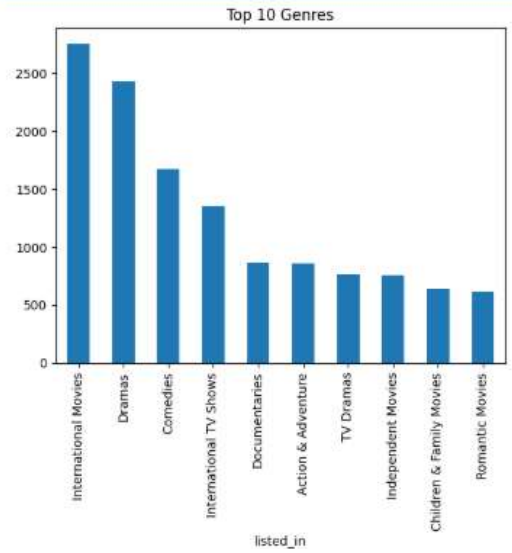


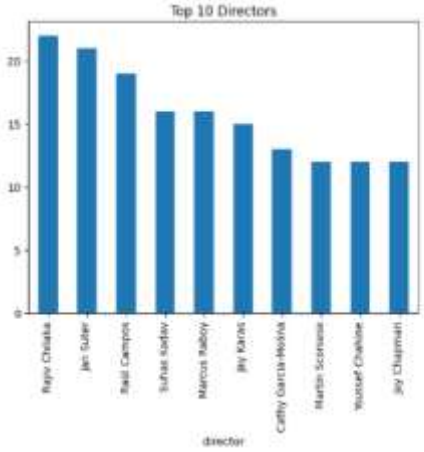

```
# Name: Shyam M (241801267)
# Section 6: Genres, Directors and Actors

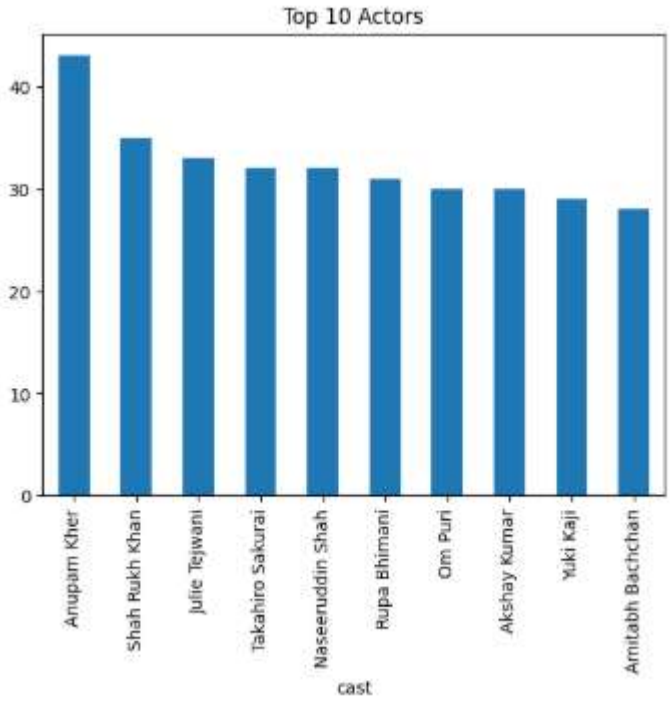
df["listed_in"].dropna().str.split(", ").explode().value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Genres")
plt.show()

df["director"].dropna().str.split(", ").explode().value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Directors")
plt.show()

df["cast"].dropna().str.split(", ").explode().value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Actors")
plt.show()
```








```
# Name: Shyam M (241801267)
# Section 7: Word Cloud of Descriptions

text = " ".join(df["description"].dropna().astype(str))

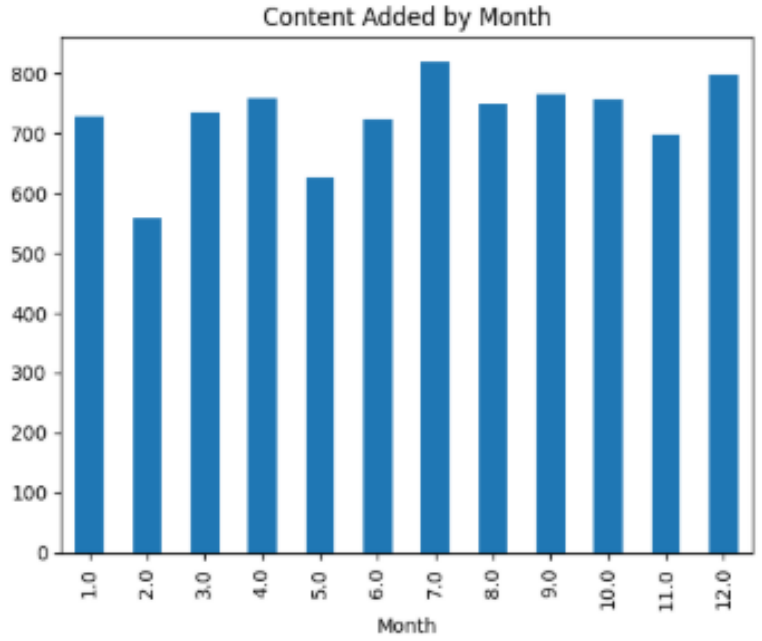
wordcloud = WordCloud(
    width=1000, height=500,
    background_color="white"
).generate(text)

plt.figure(figsize=(12, 6))
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.title("Netflix Description Word Cloud")
plt.show()
```




```
# Name: Shyam M (241801267)
# Section 8: Monthly Content Additions

df["month_added"] = df["date_added"].dt.month
df["month_added"].value_counts().sort_index().plot(kind="bar")
plt.title("Content Added by Month")
plt.xlabel("Month")
plt.show()
```

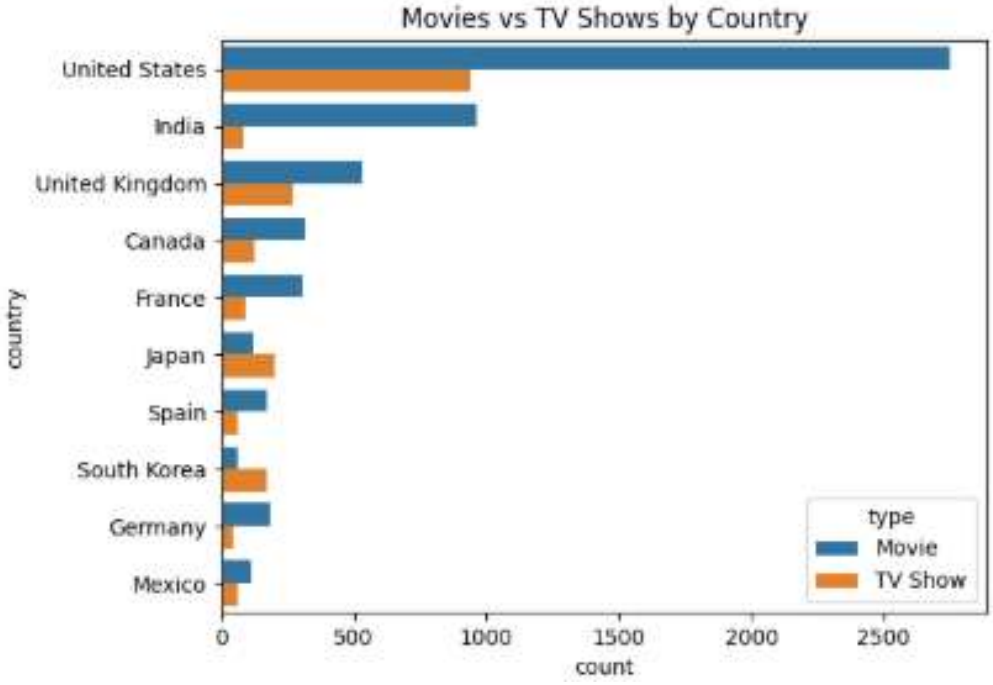


```
# Name: Shyam M (241801267)
# Section 9: Movies vs TV Shows by Country

country_type = df.dropna(subset=["country"]).copy()
country_type["country"] = country_type["country"].str.split(", ")
country_type = country_type.explode("country")

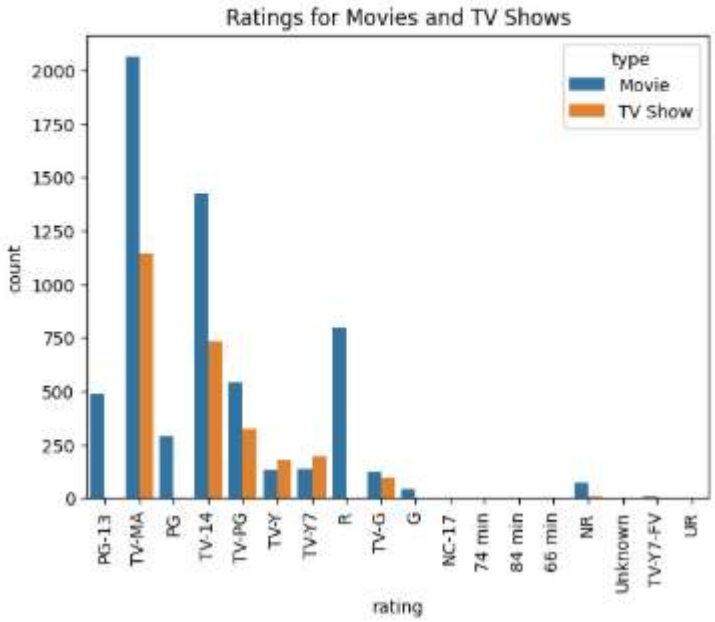
top_countries = country_type["country"].value_counts().head(10).index
country_type = country_type[country_type["country"].isin(top_countries)]

sns.countplot(data=country_type, y="country", hue="type",
              order=top_countries)
plt.title("Movies vs TV Shows by Country")
plt.show()
```



```
# Name: Shyam M (241801267)
# Section 10: Ratings by Content Type

sns.countplot(data=df, x="rating", hue="type")
plt.title("Ratings for Movies and TV Shows")
plt.xticks(rotation=90)
plt.show()
```




```
# Name: Shyam M (241801267)
# Section 11: Oldest and Newest Titles

print("Oldest Release Year:", df["release_year"].min())
print("Newest Release Year:", df["release_year"].max())

print("\nOldest Titles:\n",
      df[df["release_year"] == df["release_year"].min()][["title", "release_year"]])

print("\nNewest Titles:\n",
      df[df["release_year"] == df["release_year"].max()][["title", "release_year"]])
```

```
Oldest Release Year: 1925
Newest Release Year: 2021

Oldest Titles:
                                    title  release_year
4250  Pioneers: First Women Filmmakers*          1925

Newest Titles:
                            title  release_year
1                 Blood & Water          2021
2                     Ganglands          2021
3         Jailbirds New Orleans          2021
4                   Kota Factory          2021
5                  Midnight Mass          2021
...                          ...           ...
1468  What Happened to Mr. Cha?          2021
1551                      Hilda          2021
1696               Polly Pocket          2021
2920               Love Is Blind          2021
8437     The Netflix Afterparty          2021

[592 rows x 2 columns]
```

```
# Name: Shyam M (241801267)
# Section 12: Genre Co-occurrence Heatmap

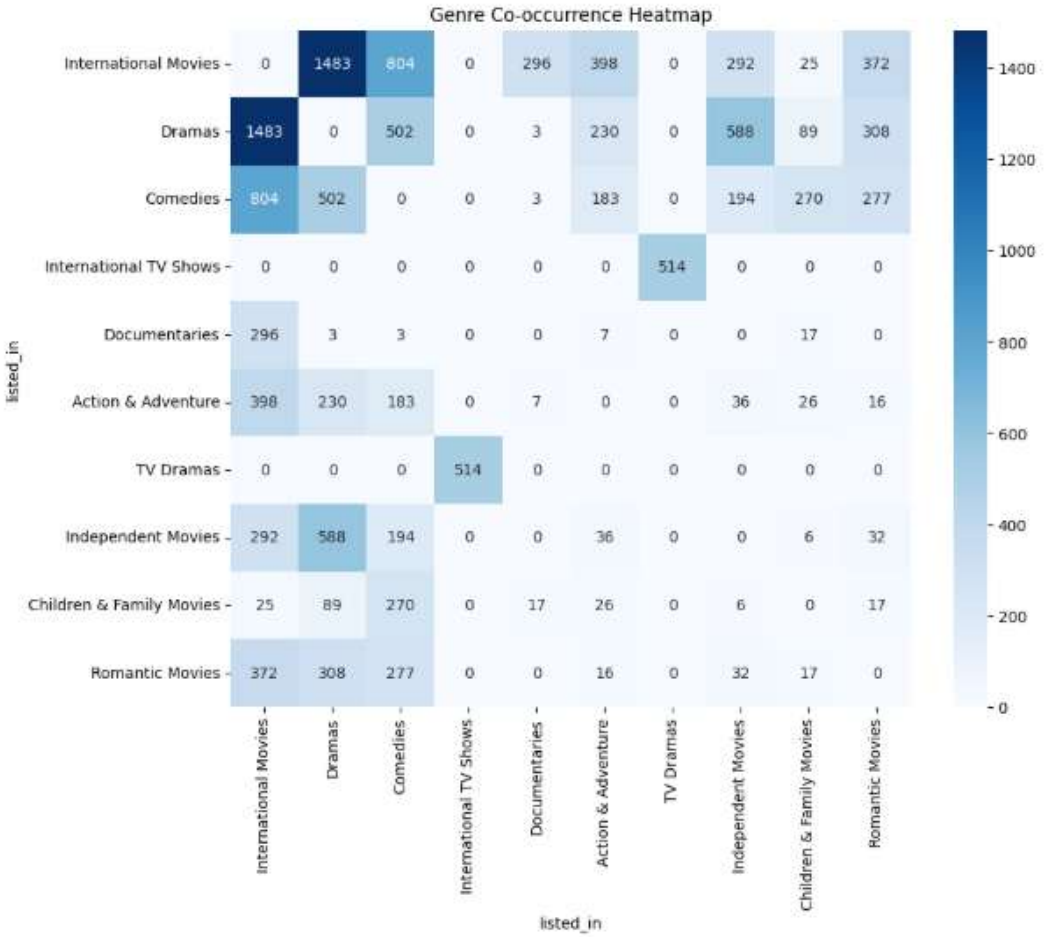
genre_pairs = Counter()

for genres in df["listed_in"].dropna():
    genre_list = [g.strip() for g in genres.split(",")]
    for a, b in combinations(sorted(set(genre_list)), 2):
        genre_pairs[(a, b)] += 1

top_genres = df["listed_in"].dropna().str.split(", ").explode().value_counts().head(10).index
matrix = pd.DataFrame(0, index=top_genres, columns=top_genres)

for (a, b), count in genre_pairs.items():
    if a in matrix.index and b in matrix.columns:
        matrix.loc[a, b] = count
        matrix.loc[b, a] = count

plt.figure(figsize=(10, 8))
sns.heatmap(matrix, annot=True, fmt="d", cmap="Blues")
plt.title("Genre Co-occurrence Heatmap")
plt.show()
```



```
# Name: Shyam M (241801267)
# Section 13: Final Summary

print("\nContent Type Distribution:\n", df["type"].value_counts())
print("\nTop 5 Genres:\n",
      df["listed_in"].dropna().str.split(", ").explode().value_counts().head())
print("\nEDA Completed Successfully.")
```

```
Content Type Distribution:
 type
Movie      6131
TV Show    2676
Name: count, dtype: int64

Top 5 Genres:
 listed_in
International Movies      2752
Dramas                    2427
Comedies                  1674
International TV Shows    1351
Documentaries              869
Name: count, dtype: int64

EDA Completed Successfully.
```

RESULT:

Thus, Exploratory Data Analysis was successfully performed on the Netflix Titles dataset using Python. The distribution of movies and TV shows, ratings, countries, release years, durations, genres, directors, and actors was analyzed. Visualizations such as count plots, histograms, bar charts, a word cloud, and a genre co-occurrence heatmap were generated to identify content distribution patterns, trends, and relationships in the dataset.